$$\begin{aligned}
\langle o|u\rangle\langle u|o\rangle &= \frac{1}{2} \\
\langle o|d\rangle\langle d|o\rangle &= \frac{1}{2} \\
\langle i|u\rangle\langle u|i\rangle &= \frac{1}{2} \\
\langle i|d\rangle\langle d|i\rangle &= \frac{1}{2}. \qquad (2.8)
\end{aligned}$$

In the first two equations, |*o*> takes the role of |*A*> from Eqs. 2.1 and 2.2. In the second two, |*i*> takes that role. These conditions state that if the spin is oriented along *y*, and is then measured along *z*, it is equally likely to be *up* or *down*. We should also expect that if the spin were measured along the *x* axis, it would be equally likely to be *right* or *left*. This leads to additional conditions:

$$\begin{aligned}
\langle o|r\rangle\langle r|o\rangle &= \frac{1}{2} \\
\langle o|l\rangle\langle l|o\rangle &= \frac{1}{2} \\
\langle i|r\rangle\langle r|i\rangle &= \frac{1}{2} \\
\langle i|l\rangle\langle l|i\rangle &= \frac{1}{2}. \qquad (2.9)
\end{aligned}$$